CRITICAL CARE MEDICINE
Copyright © 1994 by Williams & Wilkins
Vol. 22, No. 9
Printed in U.S.A.

Understanding the pressure cost of ventilation: Why does high-frequency ventilation work?

JOSÉ G. VENEGAS, PHD; JEFFREY J. FREDBERG, PHD

Objectives: **To understand when the use of high-frequency ventilation would be advantageous, we formulated the problem of achieving adequate alveolar ventilation at minimal pressure cost by dividing it into two simpler problems: a) the pressure cost per unit of convective oscillatory flow; and b) the convective flow cost necessary to achieve a unit of alveolar ventilation.**

Methods: **Simple solutions for each of these cost functions were formulated using established models of gas exchange and lung mechanics, including the effects of lung inflation tidal volume and respiratory frequency in alveolar ventilation, nonlinear lung tissue compliance, and alveolar recruitment and derecruitment. Solutions to these models were combined to assess the total pressure cost of high-frequency ventilation as a function of the ventilatory settings and the pathophysiologic variables of the patient.**

Main Results: **The model predicted that for variables applicable to an infant with respiratory distress syndrome, the selection of positive end-expiratory pressure (PEEP) becomes critical because the penalties in pressure cost are amplified for both high and low values of PEEP. The selection of frequency is not as critical for frequencies >10 Hz, although it is more important than in the normal neonatal lung.**

Conclusions: **This analysis illustrates the importance of using high-frequency ventilation in infant respiratory distress syndrome and of optimizing the amount of PEEP. It also points out the danger of barotrauma in the derecruited lung. When the lungs are in a derecruited state, the combinations of frequency, PEEP, and tidal volume that yield adequate ventilation with safe**

Table of abbreviations

C	compliance
Dalv	normalized peak alveolar distention
$D(P_c)$	normal distribution of alveolar closing pressure
f	frequency
f_c	corner frequency
f_o	resonance frequency
F	normalized frequency
F_T	transition frequency
FRC	functional residual capacity
γ	expiratory time constant of the respiratory system
I	inertance
IRDS	infant respiratory distress syndrome
N_r	fraction of alveolar units remaining recruited
N_rPEEP	fraction of alveolar units remaining recruited at a given amount of positive end-expiratory pressure after steady state
N_v	viable alveolar units of the lung
P_{alv}	alveolar pressure
$P_{alv,peak}$	peak alveolar pressure
P_c	alveolar closing pressure
P_{ca}	carinal distending pressure
ΔP_{ca}	peak to peak amplitude of oscillatory carinal distending presssure swings
Pca_{peak}	peak carinal distending pressure
σ_{PC}	standard deviation of alveolar closing pressure
P_{ca}	carinal pressure
P_{ct}	central airway distending pressure
PEEP	positive end-expiratory pressure
P_{lt}	alveolar distending pressure
P_{pl}	pleural pressure
q	quality factor
Q	flow cost of ventilation
R	resistance
Raw	airway resistance
σ	standard deviation
T_E	expiratory time
TLC	total lung capacity
$\dot{V}_A$	alveolar ventilation
V_D	sum of the anatomical and equipment deadspace
V_{EE}	end-expiratory volume
V_L	volume of a totally recruited normal lung
V_L^*	volume of recruited alveoli
V_T	tidal volume
$\|Z\|$	magnitude of lung compliance or pressure cost of flow
Z_{cw}	impedance of the chest wall

From the Department of Anesthesia (Bioengineering), Massachusetts General Hospital and Harvard Medical School, and the Department of Environmental Health, Harvard School of Public Health, Boston, MA.

This study was supported, in part, by National Institutes of Health Grant HL38267.

Address requests for reprints to: Dr. José G. Venegas, Department of Anesthesia (Bioengineering) (CLN–2), Massachusetts General Hospital, Boston, MA 02114.

0090-3493/94/2209-S49$03.00/0

distention of recruited alveoli are severely limited. (Crit Care Med 1994; 22:S49–S57)

KEY WORDS: **high-frequency ventilation; barotrauma; mechanical ventilation; pulmonary gas exchange; positive end-expiratory pressure; tidal volume; pulmonary diseases; respiratory insufficiency**

Early experiences with high-frequency ventilation indicated that adequate gas exchange was achievable with reduced tidal expansion of the lungs as respiratory frequency was increased above the physiologic range (1–4). It was expected that reduction of tidal lung expansion would diminish the risks of ventilator-related lung injury in all types of lung pathology. Over a decade later, these expectations have been realized, but only for specific types of lung pathology such as occur in infant respiratory distress syndrome (5, 6). In patients with other disorders, such as those patients with airway diseases, ventilation at high frequency has not only been ineffective but could be harmful to the lung by creating lung hyperinflation and inadvertent positive end-expiratory pressure (PEEP) (7, 8). Aside from empirically derived strategies for the use of high-frequency ventilation, there are currently no mechanistically based guidelines for a rational selection of frequency, tidal volume, and mean airway pressures during high-frequency ventilation. This article brings together fundamental aspects of high-frequency ventilation and provides a basis for understanding why and when the use of high-frequency ventilation can be advantageous. These bases should serve as a platform to develop rational ventilatory strategies during mechanical ventilation.

With the exception of pure molecular diffusion, all known mechanisms of gas transport in the airways depend on convective fluid motion. There are two aspects of fluid motion: gas transport is driven by convective fluid motion (the transport aspect) and convective fluid motion is driven by pressure differences imposed by the chest wall or by the ventilator (the mechanics aspect). Insofar as fluid motion is concerned, the pressure difference from the airway opening to the alveolus is the primary driving pressure of interest. But pressures that drive flow also distend tissues. Parenchymal tissues are distended by the pressure difference between the alveolus and the pleura and the chest wall by the pressure difference between pleural and body surfaces. The pressure distending intrathoracic airways is well approximated by the carinal distending pressure. A central question in high-frequency ventilation is how to achieve adequate amounts of gas transport while exposing airways and parenchymal tissues neither to distending pressures that are so small that airways close or alveolar units derecruit, nor so large that these tissues are damaged and their function is compromised (barotrauma). Other important questions remain but are beyond the scope of this article. These issues include, among others, the nonuniform behavior of the lung and chest wall (9–14).

The pressure distending lung tissues can be partitioned into a steady end-expiratory pressure component (PEEP), on which is added an oscillatory component associated with the imposed mechanical ventilation. If we postulate that eucapnic ventilation might be achieved by some judicious selection of oscillation frequency, tidal volume, and PEEP, then, in cases where barotrauma (too large a maximum alveolar distending pressure) and/or alveolar derecruitment (too small a PEEP) are of concern, a small increment in frequency of oscillation conceivably might allow for a small decrement in tidal volume and, thereby, a small decrement in phasic excursion of alveolar distending pressure. This expectation would be one of simultaneously lessening the potential for barotrauma and phasic derecruitment. This concept represents the first promise of high-frequency ventilation, and how it might work is elaborated below. A second promise of high-frequency ventilation, which we do not address in this paper, concerns the distribution of regional lung expansion and gas exchange. At conventional frequencies, the distribution of inspired gas is, for the most part, controlled by the distribution of regional lung compliance. Insofar as most disease processes engender nonuniform regional lung compliance, heterogeneous regional expansion and ventilation necessarily follow. High-frequency ventilation promises to change that situation because, as frequency is increased, the distribution of regional expansion depends less on regional tissue compliance (15) and more on the properties of central airways—specifically airways resistances (16), inertances (the "inertive" component of impedance), and branching angles (9, 17, 18)—and even more importantly on the mechanical properties of the chest wall (14). Thus, high-frequency ventilation promises to bring at least two additional degrees of freedom to mechanical ventilation: a) a smaller magnitude of phasic tissue distention; and b) some degree of control over its regional distribution.

PROBLEM FORMULATION

We formulated the problem of achieving adequate alveolar ventilation at minimal pressure cost by dividing it into two simpler problems by expressing the

pressure cost for achieving a unit of alveolar ventilation as the product of the pressure cost incurred per unit of convective flow times the convective flow cost necessary to achieve a unit of alveolar ventilation. Simple solutions for each of these cost functions are described below and are combined to assess the total pressure cost of high-frequency ventilation as a function of the ventilatory settings and the pathophysiologic variables of the patient.

Flow Cost of Achieving Ventilation. The gas transport aspect of the high-frequency ventilation problem can be defined as the flow cost incurred generating a unit of alveolar ventilation: $Q = (V_T f/\dot{V}_A)$, where Q is flow cost of ventilation, V_T is tidal volume, f is frequency, and $\dot{V}_A$ is alveolar ventilation. Venegas and co-workers (19) demonstrated that flow cost can be plotted against a normalized frequency: $F = f/(\dot{V}_A/V_D)$, where F is a normalized frequency and V_D is the sum of anatomical and equipment deadspace. This arrangement enabled Venegas and co-workers to define a single relationship that is valid for multiple mammalian species over a wide range of lung sizes and frequencies. This relationship (Fig. 1), which defines a transition frequency ($F_t \approx 5$) between conventional ventilation and high-frequency ventilation, has also been validated for individual animals at various levels of lung inflation (20, 21). Within the conventional ventilation domain (normalized frequency <5), the flow cost of ventilation can be expressed as $Q = 1 + F$, which can be easily derived from the well-known relation, $\dot{V}_A = (V_T - V_D) \cdot f$.

Within the high-frequency ventilation domain (normalized frequency >5), flow cost is described by the equation, $Q = (F/0.18)^{0.54}$, which, expressed in dimensional variables, gives the equation $(V_T f/\dot{V}_A) = k_2 (f/[\dot{V}_A/V_D])^{0.54}$.

Because total deadspace volume increases in proportion to end-expiratory lung volume (22), flow cost can be expressed as a function of frequency and PEEP, and can be visualized in the form of a three-dimensional surface (Fig. 2).

Flow cost increases with increasing frequency (moving along a constant PEEP line in Fig. 2), reflecting a decrease in gas transport efficiency due to the growing fraction of wasted deadspace ventilation (23). Flow cost also increases with increasing PEEP (moving along a constant frequency line in Fig. 2) because expanding the anatomical deadspace as the lung expands also results in a higher fraction of wasted deadspace ventilation (20). This increase in flow cost with PEEP at higher frequency is more pronounced than at lower frequency.

Flow cost is also highly dependent on ventilatory design and breathing circuit configuration. For example, increased equipment deadspace volume or an inefficient expired gas removal system increases rebreathing and results in greater flow cost of ventilation. Conversely, reducing equipment deadspace volume by introducing the fresh gas closer to the trachea (as in high-frequency jet ventilation using a Hi-Lo

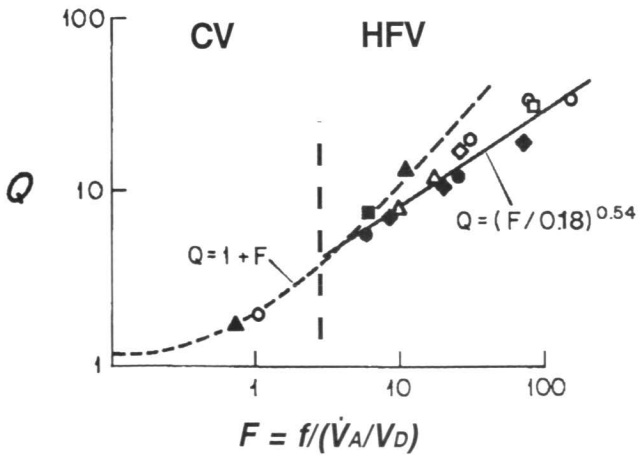


Figure 1. Oscillatory flow cost of ventilation ($Q = V_T f$), plotted vs. respiratory frequency (*F*), normalized by the ratio of alveolar ventilation ($\dot{V}_A$) to deadspace (V_D) for several mammalian species. Note the transition point where the high-frequency ventilation (*HFV*) relationship ($Q = (F/0.18)^{0.54}$) departs from the relationship for conventional ventilation (*CV*), where $Q = 1 + F$. Modified from Venegas et al (19). *Open triangles*, rats; *closed triangles*, rabbits; *closed diamonds*, monkeys; *open circles*, dogs; *open squares*, dogs; *closed circles*, dogs; *open triangles*, humans; *closed squares*, horses.

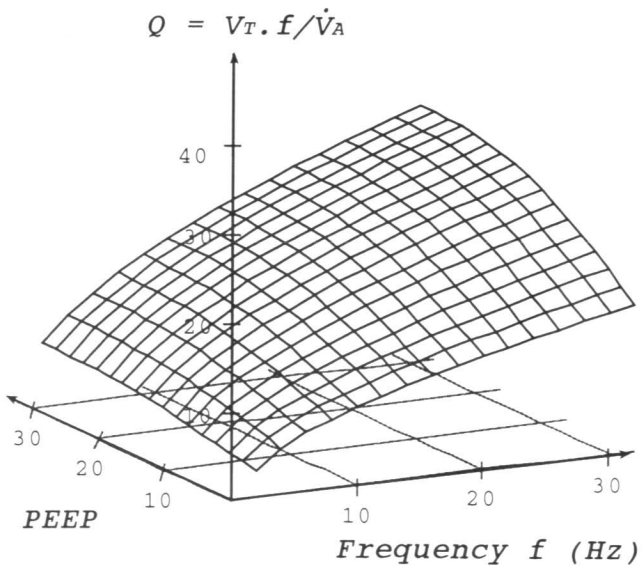


Figure 2. Surface representing the frequency (*f*) and positive end-expiratory pressure (*PEEP*) dependencies of the flow cost (*Q*) of alveolar ventilation ($\dot{V}_A$). V_T, tidal volume.

tube), results in lower flow cost for a given combination of frequency and PEEP.

Pressure Cost of Achieving Flow. We need first to define the pertinent pressure that can be related to barotrauma and alveolar derecruitment. In high-frequency ventilation, the endotracheal tube imposes a large fraction of the overall inertive and resistive load of the ventilator (24, 25). Therefore, pressure measured at the airway opening is a poor indicator of the pressure that is distending lung tissues. Central airways are distended by the pressure difference ($P_{ct} = P_{ca} - P_{pl}$), while alveolar tissues are distended by $P_{lt} = P_{alv} - P_{pl}$ (where P_{ct} is central airways distending pressure, P_{ca} is carinal pressure, P_{pl} is pleural pressure, P_{lt} is alveolar distending pressure, and P_{alv} is alveolar pressure). The phasic amplitude of alveolar distending pressure can, under certain conditions, be higher than that value measured in the proximal airway (a phenomenon called resonant amplification [26], resulting from the interaction between inertance of the airways and parenchymal elasticity). From the simplest model of lung mechanics, it can be shown that resonant amplification is present only in underdamped systems. This means that resonant amplification is only present in lungs where the quality factor (q) is greater than unity ($q > 1$). (The quality factor is defined as $q = \sqrt{I/C}/R$, where I and R are the inertance and resistance of the airways distal to the carina and C is lung compliance). As resistance increases, the resonant amplification is dampened and eliminated as the system becomes overdamped ($q < 1$). For variables representing conditions for normal infants and infants with respiratory distress syndrome measured from the carina, the model yields quality factor values of 0.32 and 1, respectively. Therefore, in the clinical pediatric situation, the system is generally either overdamped or critically damped (quality factor of 1) and carinal distending pressure swings can be expected to be always equal to or greater than the pressure swings distending the average distal alveoli. Because a large number of alveoli are located at short distances from central airways, they will generally be subjected to higher distending pressures than more distal alveoli. Thus, a safe approach in high-frequency ventilation to reduce the risk of localized tissue injury is to focus on the peak values of carinal distending pressure $(P_{ca} - P_{pl})_{peak}$. We shall examine both the pressure distending the carina as well as the pressure distending the more distal parenchymal tissues as the relevant pressure costs of ventilation.

Pressure Cost of Flow in High-Frequency Ventilation. The peak values of carinal distending pressure can be expressed as the sum of a constant positive end-expiratory distending pressure superimposed on the peak-to-peak amplitude of the oscillatory distending pressure swings (ΔP_{ca}). As a first approximation, one can estimate the peak-to-peak amplitude of the oscillatory distending pressure swings during high-frequency ventilation using the simple analog model of Figure 3, where the amplitude of the distending oscillatory pressure swing per unit flow at the carina is, by definition, the magnitude of lung impedance beyond that point. Figure 4*A* illustrates how this oscillatory pressure cost of flow rapidly decreases with increasing frequency and reaches a minimum at the resonance frequency, which can be expressed in terms of mechanical variables as $f_o = 1/(2\pi\sqrt{CI})$, where f_o is resonance frequency.

However, for overdamped physiologic systems (quality factor <1), a plateau is reached for frequencies higher than the systems' corner frequency, which is defined as $f_c = 1/(2\pi\sqrt{CR})$, where f_c is corner frequency.

Noting that the quality factor is the corner frequency divided by the resonance frequency, it follows that in overdamped systems, corner frequency occurs at lower frequency than resonance frequency. A corollary of this statement is that the greatest relative reductions in the pressure cost of flow during high-frequency ventilation are realized at frequencies lower than the corner frequency. This strong negative frequency dependence of the pressure cost of flow reflects the decreasing contributions of tissue elasticity and the increasing contribution of airways resistance to the pressure cost of flow. Thus, in respiratory distress

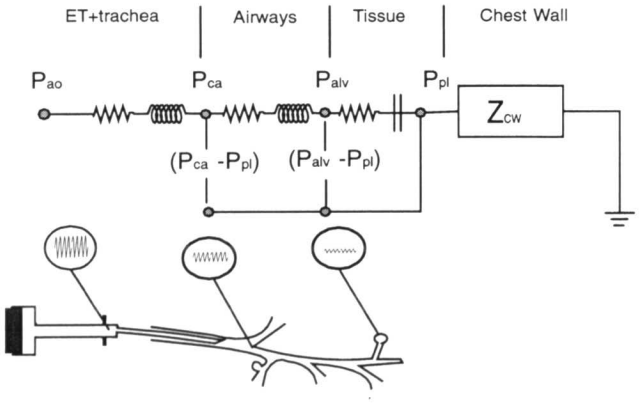


Figure 3. Model of respiratory system mechanics where pressures are represented by voltages and oscillatory flows are represented by alternating currents. Viscous losses are represented by resistive elements, while inertive and elastic loads are represented by inductors and capacitors, respectively. Distending pressures at the carina (P_{ca}–P_{pl}) and the alveoli (P_{alv}–P_{pl}) are the pressure differences between the carina and alveoli with pleural pressure. Illustrated is the damping of the pressure swings as one moves from the airway opening to the carina and to an average distal alveoli. Note that proximal alveoli could be subjected to much higher distending pressures closer to P_{ca}–P_{pl}. P_{ao}, airway opening pressure; Z_{cw}, impedance of the chest wall.

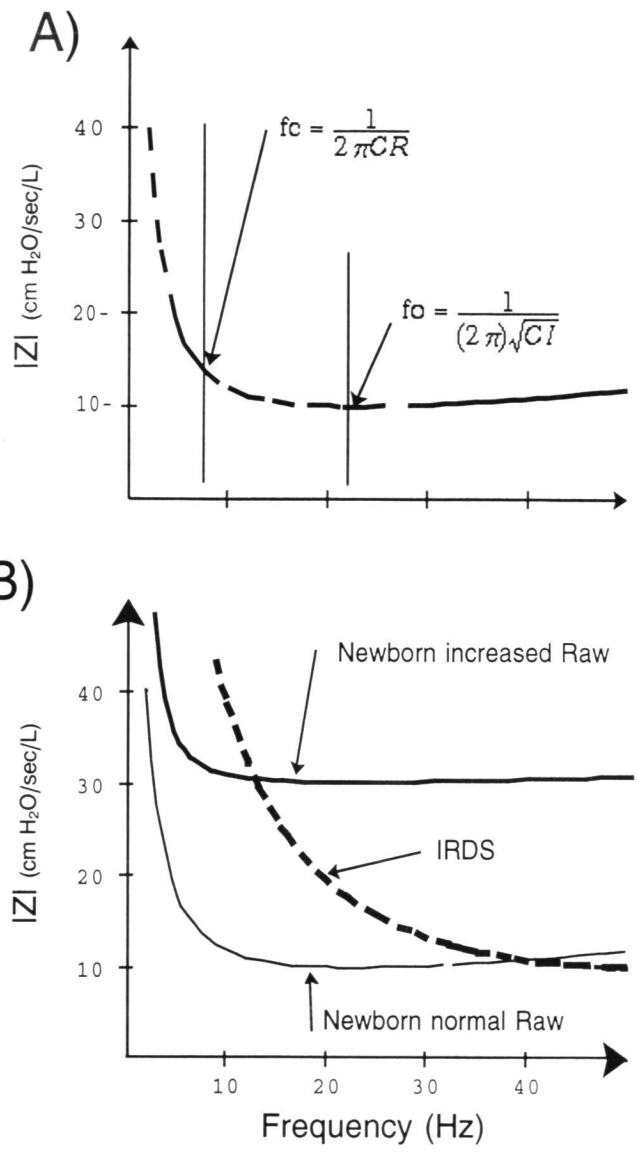


Figure 4. *A)* Plot of magnitude of lung impedance ($|Z|$) vs. frequency for typical variables of a normal newborn infant (compliance [*C*], 0.002 L/cm H_2O; resistance [*R*], 10 cm H_2O/sec/L; and inertance [*I*], 0.025 cm H_2O/sec^2/L). For these variables, the corner frequency (f_c) and resonant frequency (f_o) were 8 and 22 Hz, respectively. Note that the impedance has a minimum at $f = f_o$, but most of the decrease in this variable occurs at $f<f_c$. *B)* Plots of magnitude of lung impedance vs. frequency for normal newborn infants, infants with infant respiratory distress syndrome (*IRDS*), and normal newborns with increased airway resistance (*Raw*).

syndrome, where the lungs become stiffer and the corner frequency becomes higher, decrements in the oscillatory pressure cost of flow continue to be realized at higher frequencies than in normal lungs (Fig. 4*B*). A second corollary is that increasing resistance shifts corner frequency to lower frequencies. Thus, increasing respiratory frequency to the range of high-frequency ventilation yields diminishing returns in decreasing the oscillatory pressure cost of flow.

The inclusion of PEEP modifies the problem in important ways. At high values of PEEP, the lung becomes less compliant, thus increasing both corner frequency and resonance frequency. At moderate values of PEEP, the lung is most compliant. At low values of PEEP, the lung loses recruited alveoli and, over time, becomes stiffer. To incorporate these effects in our model, we considered two independent functions: the first function representing the nonlinear elasticity of the normal, fully recruited lung; the second function representing the effect of alveolar derecruitment on lung compliance.

Lung elasticity can be described by a relationship between lung volume (V_L) above resting functional residual capacity as a function of distending pressure ($V_L[P_{lt}]$). Such a relationship is represented by a combined function (Fig. 5*A*), including a linear portion, for a maximum alveolar distending pressure of <10 cm H_2O, and an exponential portion, for a maximum alveolar distending pressure of >10 cm H_2O:

$$V_L = P_{lt} \cdot C_l \qquad \text{for } P_{lt} \leq 10 \text{ cm } H_2O$$

$$V_L = 10 \cdot C_l \left(\frac{3}{2} - \frac{1}{2} \cdot e^{\frac{-(P_{lt} - 10)}{5}}\right) \qquad \text{for } P_{lt} > 10 \text{ cm } H_2O$$

This function has the following characteristics: a) its single variable, C_l, represents the slope of the linear portion; b) it is continuous to the first derivative; and c) V_L approaches asymptotically total lung capacity for lung distending pressures >30 cm H_2O.

Alveolar derecruitment is incorporated in the model by defining a function of PEEP ($N_r[PEEP]$) to represent the fraction of alveolar units remaining recruited at a given amount of PEEP after steady-state conditions of ventilation have been reached. To define the form of the function of PEEP, we assume that the distribution of alveolar closing pressures (P_c) obeys a normal distribution ($D[P_c]$), with a mean value and with a standard deviation (σ_{PC}). If, in the steady state, most of the derecruited alveoli remain collapsed during the breathing cycle, then the function of PEEP can be approximated to the integral of the normal distribution of alveolar closing pressures:

$$N_R(PEEP) = \int_{-\infty}^{PEEP} D(P_c)\, dP_c$$

The function $N_r(PEEP)$ therefore follows a sigmoidal curve that, at high amounts of PEEP, becomes asymptotic to the fraction of recruitable or viable alveolar units of the lung (N_v) (Fig. 5*B*).

Thus, the effect of alveolar derecruitment in the overall lung pressure-volume relationship for a given

amount of PEEP ($V_L^*[P_{lt},PEEP]$) is: $V_L^*(P_{lt},PEEP) = V_L(P_{lt}) \times N_r(PEEP)$.

For small amplitude oscillations, lung compliance can be approximated as the slope of the $V_L^*(P_{lt},PEEP)$ relationship. For larger values of tidal volume, or when the fraction of recruitable alveoli is small, this approximation may seriously overestimate lung compliance. Therefore, to calculate lung compliance for each frequency and PEEP value, we first estimated the tidal volume required for eucapnia from the flow cost vs. normalized frequency relationship. We then calculated end-expiratory volume as $V_{EE} = V_L^*(PEEP,PEEP) + V_T$ and the corresponding peak alveolar pressure ($P_{alv,peak}$). Lung compliance was finally calculated as $C = V_T/(P_{alv,peak} - PEEP)$.

Assuming, for simplicity, that resistance and inertance are insensitive to changes in lung volume, the oscillatory component of the pressure cost of flow can be represented as a function of frequency and PEEP in the form of a surface (Fig. 6).

For variables corresponding to a normal newborn (Table 1), a relatively flat pressure cost of flow surface occurs over a wide range of frequency-end expiratory pressure combinations. In this case, the pressure cost of flow becomes high only at the very low frequency (where high tidal volumes have to be used) or high PEEP, where lung compliance decreases due to overinflation, and at very low PEEP values where substantial derecruitment of alveoli also results in higher lung stiffness. For a patient with infant respiratory distress syndrome in whom there is only 40% of recruitable alveoli (Table 1) and mean alveolar closing pressure is increased to 5 cm H_2O, a plateau in the pressure cost of flow surface is no longer realized and inadequate selection of PEEP or frequency brings about substantial increases in the pressure cost of flow (Fig. 6).

Pressure Cost of Ventilation. For every combination of frequency and PEEP, the value of the pressure cost of flow on the surface of Figure 6 times the value of the flow cost of ventilation on the surface of Figure 2 plus the value of PEEP yields a new surface describing the peak distending pressure to which carinal and nearby alveolar tissues are exposed during oscillatory ventilation (Fig. 7, *top*).

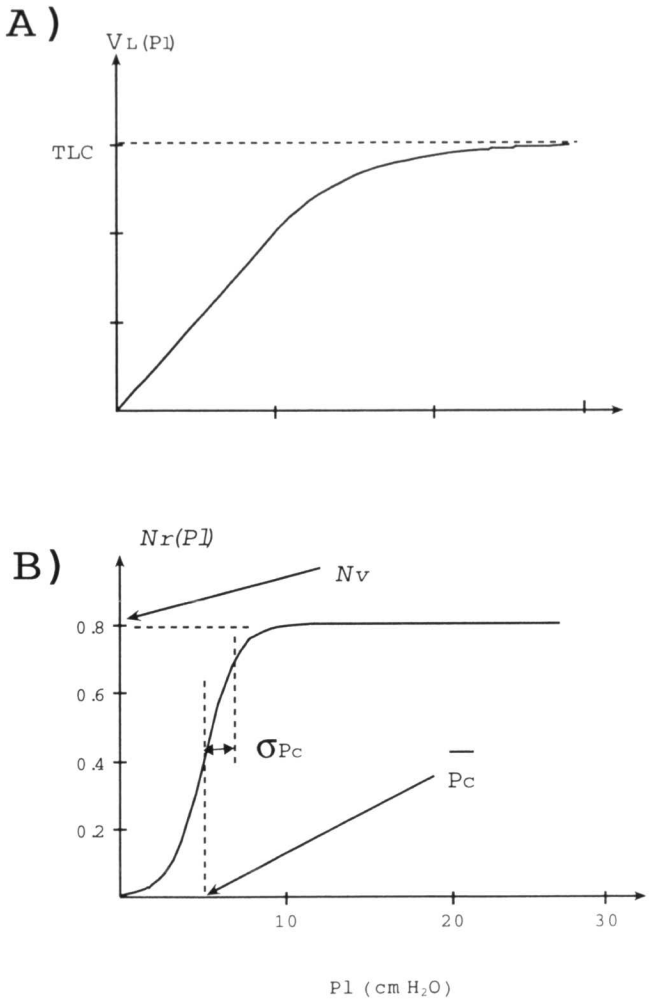


Figure 5. *A)* Idealized pressure-volume relationship for a fully recruited lung, corresponding to the equations:

$$V_L = P_{lt} \cdot C_l \qquad \text{for } P_{lt} \leq 10 \text{ cm } H_2O$$

$$V_L = 10 \cdot C_l \left(\frac{3}{2} - \frac{1}{2} \cdot e^{\frac{-(P_{lt}-10)}{5}}\right) \qquad \text{for } P_{lt} > 10 \text{ cm } H_2O$$

B) Fraction of alveolar units remaining recruited (N_r) as a function of positive end-expiratory pressure after a maximal recruitment maneuver and after a steady-state had been reached. N_v corresponds to the maximal fraction of recruitable alveoli. *VL*, lung volume; *Pl*, lung pressure; *TLC*, total lung capacity; *Pc*, alveolar closing pressure; σ, standard deviation.

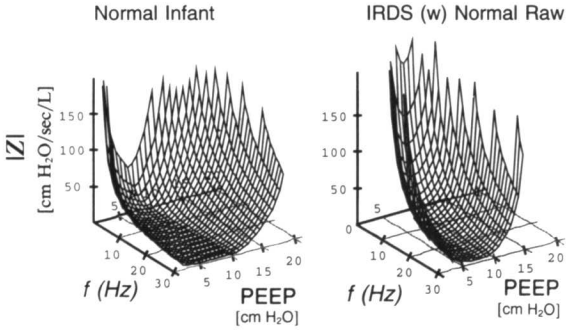


Figure 6. Surfaces of carinal oscillatory pressure cost of flow | Zca |, including the effect of changes in positive end-expiratory pressure (*PEEP*) for a normal newborn infant and one with infant respiratory distress syndrome (*IRDS*) having 40% of his alveoli recruited but with normal airway resistance (*Raw*).

Table 1. Values of variables used in the simulations

	Lung Compliance (L/cm H_2O)	Nv	Lung Resistance (cm H_2O·sec/L)	Lung Inertance (cm H_2O·sec^2/L)	Mean of Airway Closing Pressures (cm H_2O)	σ_{PC} (cm H_2O)
Normal Newborn	0.02	1	10	0.025	2	1
Infant RDS	0.0015	0.4	10	0.025	5	2
Infant RDS + high resistance	0.0015	0.4	50	0.025	5	2
Newborn + high resistance	0.02	1	50	0.025	2	1

Other variables kept constant: deadspace (2.2 mL) and alveolar ventilation (3 mL/sec).

Nv, fraction of recruitable (viable) alveoli; σ_{PC}, standard deviation of the probability distribution of airway closing pressures; RDS, respiratory distress syndrome.

For variables applicable to a normal newborn, the pressure cost of ventilation increases greatly at high PEEP values. This finding is the result of the combination of two effects. First, with increasing lung volume, the parenchyma becomes less compliant (stiffer), increasing the elastic contribution to the pressure cost of flow. Second, the anatomical deadspace also increases with lung volume, thus increasing the flow cost of alveolar ventilation. To a lesser degree, the pressure cost of ventilation also increases at reduced amounts of PEEP due to the decrease in compliance by alveolar derecruitment. For normal lung variables at intermediate amounts of PEEP, the pressure cost of ventilation is rather insensitive to changes in frequency for frequencies exceeding the corner frequency (5 Hz). In contrast, for variables corresponding to an infant respiratory distress syndrome, the effects of changing PEEP and frequency are substantially magnified. In this case, selection of PEEP becomes critical because the penalties in pressure cost are amplified for both high and low amounts of PEEP. Compared with the effects of PEEP, the selection of frequency is not as critical, at least for frequencies >10 Hz, although they are still much more important than in the normal neonatal lung. This fact is mainly because the loss in lung compliance results in a higher corner frequency, thus allowing further gains in pressure cost of flow at higher frequency.

With increased airways resistance, the pressure cost of ventilation increases substantially at high frequency because few gains in pressure cost of flow are realized above the corner frequency, while the increased flow cost of ventilation translates into increased resistive pressure losses and hence, higher peak carinal distending pressures (Fig. 7, *Bottom*).

The effect of frequency on pressure cost of ventilation can be better recognized by examining slices

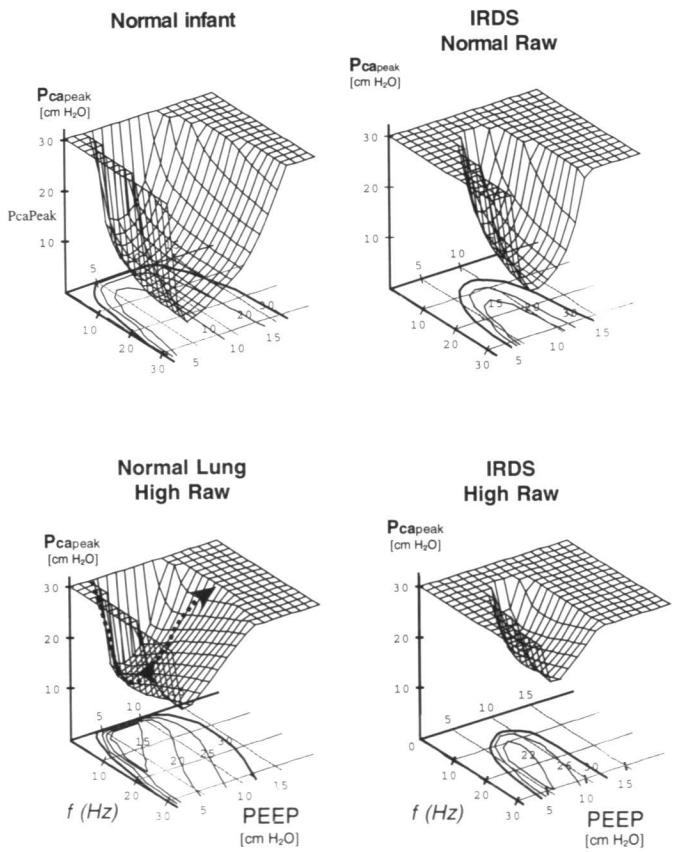


Figure 7. *Top)* Surfaces of peak carinal distending pressure (Pca_{peak}) as a function of frequency (*f*) and positive end-expiratory pressure (*PEEP*) for a normal newborn infant and one with infant respiratory distress syndrome (*IRDS*). For clarity, Pca_{peak} is arbitrarily set equal to 30 cm H_2O for f-PEEP combinations exceeding this value. *Bottom)* Surfaces and contour plots of peak carinal distending pressure as a function of frequency and positive end-expiratory pressure for increased airways resistance (*Raw*) in a newborn with normal compliance and in an infant with respiratory distress syndrome. *Dotted path*, the effect of increased mean lung volume caused by "auto-PEEP" as f is increased in high-frequency jet ventilation.

through the surfaces for the examples described above taken at a fixed value of PEEP of 10 cm H_2O (Fig. 8). Note how the optimal frequency, requiring the lowest pressure cost of ventilation, shifts to a higher frequency range as compliance is reduced and to a lower frequency range as resistance is increased.

Peak Alveolar Distention. While we focused the model to study carinal distending pressures, what are the implications for peak distention of the average alveolus? As shown in Figure 9, in the normal neonatal lung, eucapnic alveolar ventilation can be obtained

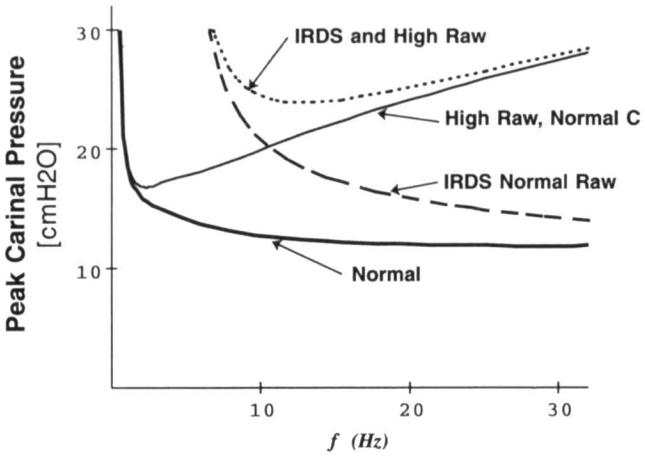


Figure 8. Peak carinal pressures vs. frequency (*f*) at a constant positive end-expiratory pressure of 10 cm H_2O for examples of a normal newborn infant and one with infant respiratory distress syndrome (*IRDS*, decreased compliance [*C*]) with normal and with increased airway resistance (*Raw*). Note that increased airway resistance shifts the point of minimal pressure cost of ventilation to a lower frequency.

$$D_{alv} = \left(V_{EE} + V_T\right) / TLC$$

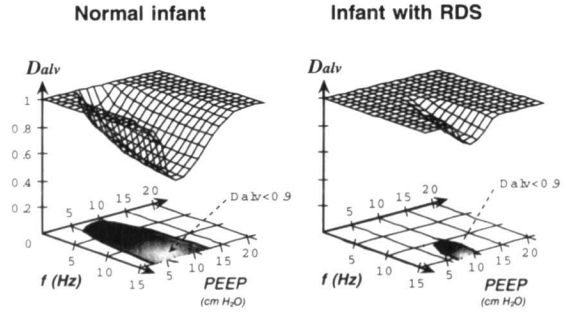


Figure 9. Surface of normalized peak alveolar distention (*Dalv*) plotted against frequency (*f*) and positive end-expiratory pressure (*PEEP*) for a normal infant and one with respiratory distress syndrome (*RDS*). Normalized peak alveolar distention is defined as lung volume at end-expiration (V_{EE}) plus tidal volume (V_T), normalized by total lung capacity (*TLC*). For clarity, normalized peak alveolar distention is arbitrarily set equal to unity for f-PEEP combinations exceeding this value. *Shaded areas*, the "safe" combinations of frequency and positive end-expiratory pressure that result in peak alveolar distention <90% of total lung capacity.

with peak alveolar distention of <90% of total lung capacity for a wide range of combinations of frequency and PEEP. Within this range, peak alveolar distention is minimized at a clearly defined optimal PEEP level. To the degree that PEEP is increased above this optimal value, peak alveolar distention will be increased, although peak distention will only reach total lung capacity at relatively large levels of PEEP. In contrast, because a low number of recruited alveoli are available to accommodate the tidal volume in the infant respiratory distress syndrome, lung peak distention encroaches on total lung capacity for all but a small range of frequency-PEEP combinations. This analysis thus illustrates the importance of using high frequencies in infant respiratory distress syndrome and of optimizing the amount of PEEP. It also points out the danger of barotrauma in the derecruited lung. When the lungs are in a derecruited state, the combinations of frequency, PEEP, and tidal volume that yield adequate ventilation with safe distention of recruited alveoli are severely limited. Thus, an important strategy to prevent barotrauma in mechanical ventilation of patients with infant respiratory distress syndrome is to recruit as many alveoli as is safely possible using periodic sustained inflations, and to keep them recruited by means of maintaining a relatively high amount of PEEP. This approach is the ventilatory strategy that has long been advocated by Froese and collaborators (27, 28).

CONCLUSIONS

We combined a simple model of lung mechanics with known gas transport data to explore some of the factors that come into play in mechanical ventilation of the neonate. While the model is an idealization, it embodies essential features thought to be most relevant. Accordingly, the qualitative nature of the predictions and strategies derived from them are expected to cross over to the clinical setting.

There are other important considerations that have not been directly addressed by this model. Among these considerations are the nonuniform distribution of ventilation in the lungs, the compliant behavior of the bronchi (29), the changes in mean lung volume created by incomplete exhalation and inadvertent PEEP (8), and the effects of changing respiratory waveform or inspiration/expiration ratio (30). Changes in mean lung volume are particularly relevant to high-frequency jet ventilation, where expiratory flow is generated exclusively by the elastic recoil of the lungs. This effect is less important in high-frequency oscillatory ventilation, where expiratory flows are aided by active suction in the part of the oscillator. In the

case of high-frequency jet ventilation, the decrease in expiratory time as frequency or inspiration/expiration ratio are increased could cause substantial gas trapping and result in a dynamic increase in end-expiratory volume (functional residual capacity). This inadvertent increase in functional residual capacity (FRC) can be described by the expression, $\Delta FRC = V_T/(e^{[T_E/\tau]} - 1)$, where T_E is the expiratory time and τ is the expiratory time constant of the respiratory system. In the clinical case of infant respiratory distress syndrome, the expiratory time constant is short ($\approx$0.12 sec). Even for frequency as high as 5 Hz and inspiration/expiration ratio of 1/2, the dynamic increase in functional residual capacity is <1/2 tidal volume. In normal or obstructive lungs, if the ventilator is kept at the same settings, it may result in a stacking of several tidal volumes inside the lungs before a steady-state lung volume is reached. In the illustration of the surface (Fig. 8), the effect of this dynamic increase in functional residual capacity can be visualized as a migration to higher and higher iso-PEEP lines as frequency is increased. The main consequence of this effect is a premature increase in mean and peak distending pressures with increasing frequency. Dynamic gas trapping can also occur, although to a lesser extent, in high-frequency oscillatory ventilation at high oscillatory flow rates, when expiratory flow becomes limited by airway collapse (8, 31). These considerations provide a further justification for selecting lower values of frequency in cases of increased airways resistance; great caution must be followed for signs of hyperinflation.

REFERENCES

1. Lunkenheimer PP, Rafflenbeul W, Keller O, et al: Application of transtracheal pressure oscillations as a modification of "diffusion respiration." *Br J Anaesth* 1972; 44:627
2. Klain M, Smith RB: High-frequency percutaneous transtracheal jet ventilation. *Crit Care Med* 1977; 5:280–287
3. Bohn DJ, Miyasaka K, Marchak BE, et al: Ventilation by high-frequency oscillation. *J Appl Physiol* 1980; 48:710–716
4. Sjostrand U: High-frequency positive-pressure ventilation (HFPPV): A review. *Crit Care Med* 1980; 8:345–364
5. Keszler M, Donn SM, Bucciarell RL, et al: Multicenter controlled trial comparing high-frequency jet ventilation and conventional mechanical ventilation in newborn infants with pulmonary interstitial emphysema. *J Pediatr* 1991; 119:85–93
6. Clark RH, Gerstmann DR, Null DM, et al: Pulmonary interstitial emphysema treated by high-frequency oscillatory ventilation. *Crit Care Med* 1986; 14:926–930
7. Saari AF, Rossing TH, Solway J, et al: Lung inflation during high-frequency ventilation. *Am Rev Respir Dis* 1984; 129:333–336
8. Simon BA, Weinmann GG, Mitzner W: Mean airway pressure and alveolar pressure during high-frequency ventilation. *J Appl Physiol* 1984; 57:1069–1078
9. Allen JL, Frantz ID III, Fredberg JJ: Regional alveolar pressure during periodic flow: Dual manifestations of gas inertia. *J Clin Invest* 1985; 76:620–629
10. Boynton BR, Fredberg JJ, Buckley BG, et al: Rib cage versus abdominal displacement in rabbits during forced oscillations to 30 Hz. *J Appl Physiol* 1987; 63:309–314
11. Venegas JG, Yamada Y, Custer J, et al: Effects of respiratory variables on regional gas transport during high-frequency ventilation. *J Appl Physiol* 1988; 64:2108–2118
12. Yamada Y, Venegas JG, Burnham C, et al: Regional mapping of gas transport during high-frequency and conventional ventilation. *J Appl Physiol* 1989; 66:1209–1218
13. Venegas JG, Yamada Y, Burnham C, et al: Local gas transport in eucapnic ventilation: Effects of gravity and breathing frequency. *J Appl Physiol* 1990; 68:2287–2295
14. Venegas JG, Tsuzaki K, Fox BJ, et al: Regional coupling between chest wall and lung expansion during HFV: A positron imaging study. *J Appl Physiol* 1993; 74:2242–2252
15. Tsuzaki K, Hales CA, Strieder DJ, et al: Regional lung mechanics and gas transport in lungs with inhomogeneous compliance. *J Appl Physiol* 1993; 75:206–216
16. Simon BA, Tsuzaki K, Venegas JG: Ventilation distribution and regional impedance during partial unilateral bronchial obstruction. *J Appl Physiol*, In Press
17. Tsuda A, Kamm RD, Fredberg JJ: Periodic flow at airway bifurcations. Part II: Flow partitioning. *J Appl Physiol* 1990; 69:553–561
18. Tsuda A, Fredberg JJ: Periodic flow at airway bifurcations. Part I: Development of steady pressure differences. *J Appl Physiol* 1990; 69:546–552
19. Venegas JG, Hales CA, Strieder DJ: A general dimensionless equation of gas transport by high-frequency ventilation. *J Appl Physiol* 1986; 60:1025–1030
20. Yamada Y, Venegas JG, Strieder D, et al: Effects of mean airway pressure on gas transport during high frequency ventilation in dogs. *J Appl Physiol* 1986; 61:1896–1902
21. Kamitsuka MD, Boynton BR, Villanueva D, et al: Frequency, tidal volume, and mean airway pressure combinations that provide adequate gas exchange and low alveolar pressure during high frequency oscillatory ventilation in rabbits. *Pediatr Res* 1990; 27:64–69
22. Hughes JMB, Hoppin FG, Mead J: Effect of lung inflation on bronchial length and diameter in excised lungs. *J Appl Physiol* 1972; 32:25–35
23. Venegas JG, Custer J, Kamm RD, et al: A relationship for gas transport during high-frequency ventilation in dogs. *J Appl Physiol* 1985; 59:1539–1547
24. Gavriely N, Solway J, Loring SH, et al: Pressure-flow relationships of endotracheal tubes during high-frequency ventilation. *J Appl Physiol* 1985; 59:3–11
25. Tsuzaki K, Simon BA, Strieder DJ, et al: Respiratory system impedance measured from the airway opening (ao) and the carina (c). Abstr. *FASEB J* 1991; (Suppl)A4427
26. Allen JL, Fredberg JJ, Keefe DH, et al: Alveolar pressure magnitude and asynchrony during high-frequency oscillations of excised rabbit lungs. *Am Rev Respir Dis* 1985; 132:343–349
27. Byford LJ, Finkler JH, Froese AB: Lung volume recruitment during high-frequency oscillation in atelectasis-prone rabbits. *J Appl Physiol* 1988; 64:1607–1614
28. Froese AB: Treatment of respiratory failure in preterm infants. *N Engl J Med* 1989; 30:1694–1695
29. Gavriely N, Solway J, Drazen JM, et al: Radiographic visualization of airway wall movement during oscillatory flow in dogs. *J Appl Physiol* 1985; 58:645–677
30. Yamada Y, Hales CA, Venegas JG: Inspiratory-to-expiratory time ratio and alveolar ventilation during high frequency ventilation in dogs. *J Appl Physiol* 1986; 61:1903–1907
31. Solway J, Rossing TH, Saari AF, et al: Expiratory flow limitation and dynamic pulmonary hyperinflation during high-frequency ventilation. *J Appl Physiol* 1986; 60:2071–2078